ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 4 SET B SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. A D flip-flop combined with some additional combinational logic is shown in Figure 1. Construct the truth table for the resulting flip-flop, analyzing all possible input combinations. Then, by using process of elimination, determine which standard flip-flop the circuit's behavior most closely resembles. 10 (CO5) (PO3)

   Hint: Consider each case individually and observe how the output changes in response to different inputs and previous states.

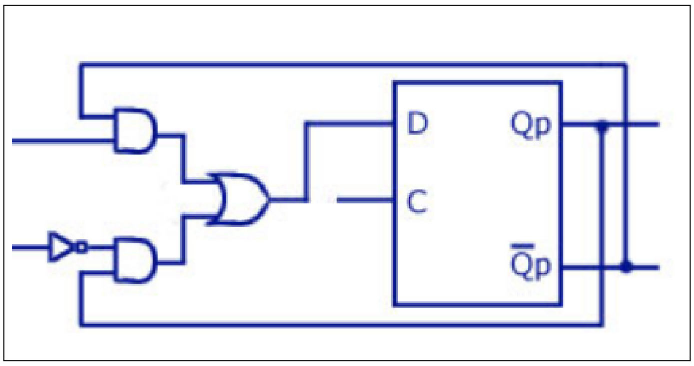

Figure 1: An unknown FlipFlop

Solution:

Step 1: Input Combinations and State Transitions

We examine all combinations of the two inputs (J, K) and how the output changes relative to the previous state (Q_n).

$$D = J \cdot \overline{Q} + \overline{K} \cdot Q$$

This is the standard logic equation used to implement JK flip-flop behavior using a D flip-flop.

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Interpretation:

- When $(J, K) = (0, 0)$, the output **remains unchanged**.
- When $(0, 1)$, the output becomes 0 **Reset**.
- When $(1, 0)$, the output becomes 1 **Set**.
- When $(1, 1)$, the output toggles:
  - $Q_n = 0 \Rightarrow Q_{n+1} = 1$
  - $Q_n = 1 \Rightarrow Q_{n+1} = 0$

Step 2: Compare Behavior with Known Flip-Flops

We now compare the observed behavior with known flip-flop types to identify the correct one.

1. D Flip-Flop

- Has only **one input** (D).
- Characteristic equation: $Q_{n+1} = D$.
- Our circuit has **two inputs**, and D is a function of J, K, Q_n.
- **Conclusion:** This is not a basic D flip-flop we are using D flip-flop to implement another behavior.

2. T Flip-Flop

- Has only **one input** (T).
- Behavior:
  - $T = 0 \Rightarrow Q_{n+1} = Q_n$ (hold),
  - $T = 1 \Rightarrow Q_{n+1} = \overline{Q_n}$ (toggle).
- Our circuit toggles only when $J = 1, K = 1$, not for a single input.
- **Conclusion:** Not a T flip-flop.

3. SR Flip-Flop

- Has two inputs: S (Set) and R (Reset).
- Behavior:
  - $(0, 0)$ Hold
  - $(0, 1)$ Reset
  - $(1, 0)$ Set

- (1, 1) Invalid

- Our circuit defines valid behavior for (1, 1), which causes toggling.
- **Conclusion:** Not an SR flip-flop.

4. JK Flip-Flop

- Has two inputs: J and K.
- Truth table:

J	K	Q_{n+1}
0	0	Q_n (No change)
0	1	0 (Reset)
1	0	1 (Set)
1	1	$\overline{Q_n}$ (Toggle)

- Matches observed behavior:
  - (0, 0) Hold
  - (0, 1) Reset
  - (1, 0) Set
  - (1, 1) Toggle
- **Conclusion:** The behavior matches that of a JK flip-flop.

Therefore, the circuit implements a JK flip-flop using a D flip-flop with logic on its input.

Any equivalent logical deduction based on the same equation will also be accepted, provided it is supported by correct reasoning.

Rubrics:

- 2 points each (times 4) for correctly identifying each case.
- 2 points for correctly identifying the resultant flip-flop's behavior with proper justification.

2. Design a 5-bit Parallel-In Serial-Out (PISO) shift register **(shifting right to left)** with a control line labeled $\text{LOAD}\backslash\overline{\text{SHIFT}}$. **Explain the working procedure of the circuit in different modes.** 5 + 5 (CO5) (PO3)

Solution:

Working Procedure and Mode Justification

The 5-bit Parallel-In Serial-Out (PISO) shift register operates in two distinct modes based on the control line labeled $\overline{\text{SHIFT}}/\text{LOAD}$. The behavior of the circuit in each mode is as follows:

- **Mode 1: Shift Mode**
  When the control line is **low** ($\overline{\text{SHIFT}} = 0$), the circuit enters **serial shift mode**. In this mode:
  - The parallel inputs are disabled.
  - On each clock pulse, the data stored in the flip-flops shifts one bit to the left (i.e., from right to left).
  - The leftmost bit (e.g., D_4) is shifted out to the serial output line.
  - A logic 0 or a new serial input (if available) may be shifted into the rightmost flip-flop.
- **Mode 2: Load Mode**
  When the control line is **high** ($\text{LOAD} = 1$), the circuit enters **parallel load mode**. In this mode:
  - The parallel input lines (D_4 to D_0) are enabled.
  - Each flip-flop in the register captures its corresponding input bit on the rising edge of the clock.
  - No shifting occurs during this mode.

Summary: The control line $\overline{\text{SHIFT}}\backslash\text{LOAD}$ determines whether the register is in load mode or shift mode. When the line is HIGH, the register loads parallel data. When the line is LOW, the register shifts its stored data leftward, outputting one bit per clock cycle starting from the MSB.

Rubrics:

- 1 point for correctly showing the modes (2).
- 1 point for correctly using D flip-flop (or any other with proper justification) (5).
- 2 points for correctly using the AND gates (10).
- 1 point for correctly using the OR gates (5).
- 2.5 points for correctly explaining the working mechanism of each mode (times 2).